

a Each validation target supports one claim, not the next

● Validation target	✓ Strongest claim it supports	✗ Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	representation capacity and technical error in the simulated regime	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan-rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Pre-specify, then escalate the reference stage by stage

Resources (Table 4)		
✓	Pre-specify the measurement contract estimand · reference · tolerance · data splits	
1	Stage 1 · Known motion vary width, extent, amplitude, timing, resolution and noise	STRAUS 3 templates × 6 states MRXCAT2.0 programmable generator CMAC dynamic phantom
2	Stage 2 · Paired material reference cine with DENSE or tagging in one session; registered; reference blinded	Stanford 51/55 · not file-verified CMAC 15 volunteers, 12 landmarks
3	Stage 3 · Tissue and decisions LGE, responsiveness, decision impact	Clinical cohorts LGE or outcomes
Counts are resource units, not independent test participants.		

c Same mean, focal deficit

